Mikhwah General Hospital
AL-BAHA CLUSTER

DPP	DEPARTMENT POLICY AND PROCEDURE		
Title	MANAGEMENT OF HYPOGLYCEMIA IN CHILDREN		
Department	Pediatric Intensive Care Unit		
Applies to	All PICU Medical and Nursing Staff		
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1.0 PURPOSE

1.1. For safe & standardized medical guidelines to treat hypoglycemia in pediatric age group.

2.0 DEFINITION/ABBREVIATION

2.1. **Hypoglycemia:** For infants and children: plasma glucose or 60 mg/dl measured by accurate laboratory method.

3.0 RESPONSIBILITY

3.1 **PICU consultant/specialist** in-charge of the unit responsible for the medical care of the patients

3.2 **Head Nurse:** Responsible in ensuring that all staffs are competent in handling procedure and in monitoring the actual performance of the staff nurse.

3.3 **Staff nurse:** Responsible in carrying out doctor's order and monitoring patient's condition.

4.0 POLICY

4.1. Patients must be stabilized in ER before shifting to PICU.

4.2. Recognition and prompt treatment of complications is expected.

4.3. In the ward: regular monitoring, RBS adjusting insulin dose, intensive health education (use of glucometer, insulin: dose calculation injection)

4.4. Before discharge, ensure good health education about how to use glucometer, how to inject insulin, management of hypoglycemia.

5.0 PROCEDURE (S)

5.1. Doctors and nurses know how hypoglycemia presents in newborn and children

5.1.1. newborn; apnea, cyanosis, poor feeding, sweating tremors, jitteriness, irritability, convulsion, coma.

5.1.2. Infant and children; Irritability, dizziness, blurred vision, difficulty speech, headache, confusion, feeling faint, personality changes, Palpitations. Hunger, Nausea Sweating, Anxiety Tremulousness Nervousness Ataxia, drowsiness, convulsion, coma.

5.2. Correct Diagnosis by physician: Reliable blood sugar estimation during symptoms.

5.3. Assessment for hypoglycemia should include:

5.3.1. History

5.3.1.1. Neonate; prematurity, IUGR, maternal diabetes, perinatal stress e.g. birth asphyxia, sepsis, shock, fasting hypoglycemia or hypoglycemia after meal (hyperinsulinemia).

5.3.1.2. Diabetes Mellitus, renal insufficiency/failure, hepatic cirrhosis/failure, other endocrine diseases, or recent surgery.

5.3.1.3. Drug Hx: Insulin usage or ingestion of an oral hypoglycemic agent, toxic ingestion, new medications

5.3.1.4. The social history may include intake and nutritional deficiency

5.3.1.5. Review systems for: weight reduction, nausea, vomiting, headache, etc...

5.3.1.6. Look for other symptoms suggesting infection

5.4. Examination:

5.4.1. Physical findings are nonspecific in hypoglycemia and generally are related to the central and autonomic nervous systems.

5.4.2. Height, weight (thin stature- ketotic hypoglycemia.

5.4.3. Assess vital signs for hypothermia, tachypnea, tachycardia, hypertension, and bradycardia (neonates)

5.4.4. The head, eyes, ears, nose, and throat examination may indicate blurred vision, pupils normal to fixed and dilated, icterus (usually cholestatic due to hepatic disease)

5.4.5. Cardiovascular disturbances may include tachycardia (bradycardia in neonates), hypertension or hypotension, and dysrhythmias.

5.4.6. Midline congenital anomalies: Ambiguous genitalia micropenis, facial anomalies, cleft palate, central incisor, nystagmus, etc.

5.4.7. Respiratory disturbances may include dyspnea, tachypnea,

5.4.8. GI disturbances may include nausea and vomiting, dyspepsia, and abdominal cramping Hepatomegaly - glycogen storage disease. Any abdominal mass

5.4.9. Skin may be diaphoretic and warm or show signs of dehydration with decrease in turgor, hyper pigmentations

5.4.10. Neurologic conditions include coma, confusion, fatigue, loss of coordination, tremors, convulsions, and diplopia and Cataract in case of galactocaemia.

5.5. Investigations

5.5.1. Obtain 5-10 ml blood red top and 20 ML urine for further investigation to be safe for 48-72 hr

5.5.2. Critical Sample: Acute Blood and Urine Tests at time of Hypoglycemia

5.6. Blood

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5.6.1. Chemistry panel with bicarbonate

5.6.2. Insulin, C-peptide

5.6.3. Cortisol, growth hormone

5.6.4. Free fatty acids, ~-hydroxy butyrate, acetoacetate

5.6.5. Lactate, ammonia

5.6.6. Total and free camitine

5.6.7. Acyl camitine profile\saves serum tube

5.7. Urine

5.7.1. Ketotic hypoglycemia

5.7.2. It is the most common cause of childhood hypoglycemia

5.7.3. It usually presents between the age of 18months and 5 years and remits spontaneously by 8 to 9 years of age

5.7.4. Hypoglycemia usually occurs during illness when food intake is limited

5.7.5. Diagnosis is confirmed by fasting for several hours under close supervision and collecting blood samples for testing At the time of hypoglycemia, there is associated ketonemia & ketonuria S.alanine levels are reduced

5.7.6. Treatment consists of frequent feedings of a high-protein, high carbohydrate diet.

5.8. Management Plan

5.8.1. For conscious and asymptomatic

5.8.1.1. Send blood sample for glucose

5.8.1.2. Offer feed or sugary drink or older children

5.8.1.3. Recheck blood sugar by 15 mins

5.8.2. Impair consciousness, convulsions or coma

5.8.2.1. Collect sample of blood for investigations

5.8.2.2. IV glucose 10% dextrose normal saline, 5ml/kg over 3-4 minute followed immediately by D10NS in maintenance IVF (6-9 ml/kg/minute). Repeat blood sugar after 15 minutes, if there is no response, repeat D10% dextrose bolus.

5.8.2.3. Increase rate and concentration of glucose infusion. If >12.5% of glucose infusion is needed to maintain normal blood sugar insertion of central line is advisable.

5.8.2.4. If still there is no response, glucagon <20 kg: 0.5 mg SC/IM/IV; may repeat in 15 minutes if necessary.

5.8.2.5. Repeat blood sugar every 15 minutes once blood sugar is controlled, oral feeding should be started. Glucose infusion should be gradually discontinued.

5.8.2.6. Try to find the cause of hypoglycemia

5.9. Documentation: history and physical examination of patient by physician, or other included services.

6.0 EQUIPMENTS

6.1. Glucometer

6.2. Gluco strip

6.3. IV fluids

7.0 ATTACHMENT/FORMS

7.1. N/A

8.0 REFERENCES

8.1. King Fahad Hospital, Policy and Procedure for Management of Hypoglycemia in Children, Policy Number: MOHIC-PICU 4/3 (4)

9.0 REVISION CLAUSE

9.1. N/A

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